

[Process management Based practical]

1. Adam is working in an IT company. He has been given a task to reduce the load of a system by killing some of the processes running in the LINUX operating system. Which commands will he use to complete the given task with the help of the following operation? • Kill processes by name

```
Chanchal@DESKTOP-GARLJAI MSYS -  
# pacman -S gcc  
resolving dependencies...  
looking for conflicting packages...  
  
Packages (8) binutils-2.45.1-1 isl-0.27-1 mpc-1.3.1-1 msys2-runtime-devel-3.6.5-1 msys2-w32api-headers-13.0.0.r0.gde62c283f-1  
msys2-w32api-runtime-13.0.0.r0.gde62c283f-1 windows-default-manifest-6.4-2 gcc-15.2.0-1  
  
Total Download Size: 72.31 MiB  
Total Installed Size: 442.30 MiB  
  
:: Proceed with installation? [y/n] y  
:: Retrieving packages...  
msys2-w32api-runtime-13.0.0.r0.gde62c283f-1-x86_64 2.9 MiB 119 KiB/s 00:25 [#####] 100%  
binutils-2.45.1-1-x86_64 5.8 MiB 149 KiB/s 00:40 [#####] 100%  
mpc-1.3.1-1-x86_64 78.5 KiB 68.0 KiB/s 00:01 [#####] 100%  
windows-default-manifest-6.4-2-x86_64 3.0 KiB 571 B/s 00:05 [#####] 100%  
msys2-runtime-devel-3.6.5-1-x86_64 5.6 MiB 118 KiB/s 00:49 [#####] 100%  
isl-0.27-1-x86_64 718.8 KiB 63.7 KiB/s 00:11 [#####] 100%  
msys2-w32api-headers-13.0.0.r0.gde62c283f-1-x86_64 5.7 MiB 88.5 KiB/s 01:06 [#####] 100%  
gcc-15.2.0-1-x86_64 51.5 MiB 277 KiB/s 03:10 [#####] 100%  
Total (8/8) 72.3 MiB 388 KiB/s 03:11 [#####] 100%  
error: failed retrieving file 'isl-0.27-1-x86_64.pkg.tar.zst' from mirror.msys2.org : operation too slow, Less than 1 bytes/sec transferred the last 10 seconds  
(8/8) checking keys in keyring [#####] 100%  
(8/8) checking package integrity [#####] 100%  
(8/8) loading package files [#####] 100%  
(8/8) checking for file conflicts [#####] 100%  
(8/8) checking available disk space [#####] 100%  
:: Processing package changes...  
(1/8) installing binutils [#####] 100%  
(2/8) installing isl [#####] 100%  
(3/8) installing mpc [#####] 100%  
(4/8) installing msys2-runtime-devel [#####] 100%  
(5/8) installing msys2-w32api-headers [#####] 100%  
(6/8) installing msys2-w32api-runtime [#####] 100%  
(7/8) installing windows-default-manifest [#####] 100%  
(8/8) installing gcc [#####] 100%  
:: Running post-transaction hooks...  
(1/1) Updating the info directory file...  
  
Chanchal@DESKTOP-GARLJAI MSYS -  
# gcc --version  
gcc (GCC) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO
```

• Kill a process based on the process name

```
Chanchal@DESKTOP-GARLJAI MSYS -  
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/  
  
Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# sleep 400  
Terminated sleep 400  
  
Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# AC  
  
Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# |  
  
Chanchal@DESKTOP-GARLJAI MSYS -  
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/  
  
Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# ps aux | grep sleep  
  
Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# |
```

• Kill a single process at a time with the given process ID

```
chancha1@DESKTOP-GARL3AI MSYS -  
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/  
chancha1@DESKTOP-GARL3AI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# sleep 400  
terminated sleep 400  
chancha1@DESKTOP-GARL3AI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# ^C  
chancha1@DESKTOP-GARL3AI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# !  
chancha1@DESKTOP-GARL3AI MSYS -  
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/  
chancha1@DESKTOP-GARL3AI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# ps aux | grep sleep  
chancha1@DESKTOP-GARL3AI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003  
# !
```

2. Write a program for process creation using C

● Orphan Process

```
GNU nano 8.7  
#include <stdio.h>  
#include <unistd.h>  
  
int main() {  
    int pid = fork();  
  
    if (pid > 0) {  
        printf("Parent Process ID: %d\n", getpid());  
        sleep(2);  
    } else {  
        sleep(5);  
        printf("Child Process ID: %d\n", getpid());  
        printf("Parent ID of Child: %d\n", getppid());  
    }  
    return 0;  
}
```

```
adity@LAPTOP-34Q2PFPC MSYS ~  
$ gcc --version  
gcc (GCC) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
adity@LAPTOP-34Q2PFPC MSYS ~  
$ nano orphan.c  
  
adity@LAPTOP-34Q2PFPC MSYS ~  
$ gcc orphan.c -o orphan  
  
adity@LAPTOP-34Q2PFPC MSYS ~  
$ ./orphan  
Parent Process ID: 1236  
  
adity@LAPTOP-34Q2PFPC MSYS ~  
$  
Child Process ID: 1237  
Parent ID of Child: 1
```

●Zombie Process

```
M ~
GNU nano 8.7
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main() {
    int pid = fork();

    if (pid > 0) {
        sleep(10); // Parent sleeps
        printf("Parent process running\n");
    } else {
        printf("Child process exiting\n");
    }
    return 0;
}
```

```
adity@LAPTOP-34Q2PFPC MSYS ~
$ nano zombie.c

adity@LAPTOP-34Q2PFPC MSYS ~
$ gcc zombie.c -o zombie

adity@LAPTOP-34Q2PFPC MSYS ~
$ ./zombie
Child process exiting
Parent process running
```

3. Create the process using fork () system call.

- Child Process creation
- Parent process creation
- PPID and PID

```
M ~  
GNU nano 8.7  
#include <stdio.h>  
#include <unistd.h>  
  
int main() {  
    int pid = fork();  
  
    if (pid == 0) {  
        // Child process  
        printf("Child Process\n");  
        printf("PID: %d\n", getpid());  
        printf("PPID: %d\n", getppid());  
    } else {  
        // Parent process  
        printf("Parent Process\n");  
        printf("PID: %d\n", getpid());  
        printf("Child PID: %d\n", pid);  
    }  
    return 0;  
}
```

```
adity@LAPTOP-34Q2PFPC MSYS ~  
$ nano 3.c  
  
adity@LAPTOP-34Q2PFPC MSYS ~  
$ gcc 3.c -o 3  
  
adity@LAPTOP-34Q2PFPC MSYS ~  
$ ./3  
Parent Process  
PID: 2193  
Child PID: 2194  
Child Process  
PID: 2194  
PPID: 2193
```